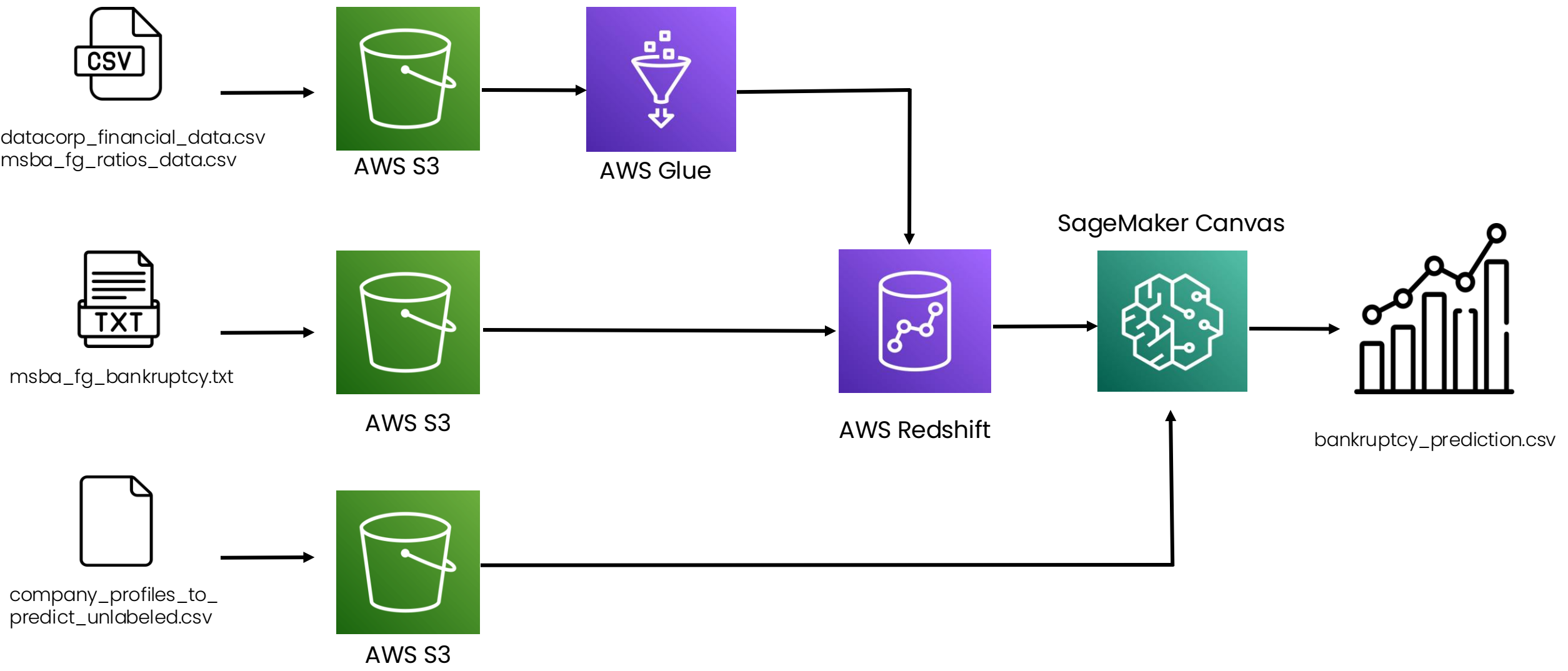


BUILDING A CLOUD-NATIVE DATA ARCHITECTURE

For MSBA Financial Group







Data Lake: AWS S3

Company ID	Net worth to Assets	Retained Earnings to Total Assets	Working Capital to Total Assets	Working Capital to Equity	Equity to Long-term Liability	Current Liabilities to Equity	Liability to Equity	Current Liability to Current Assets
MS0001	0.762423739	0.902024771	0.672770202	0.757274602	0.13804488	0.336077007	0.290021893	0.118026077
MS0002	0.801080284	0.911080284	0.711107029	0.739170289	0.10916136	0.300474018	0.10041008	0.104770282
MS0003	0.761484204	0.909602082	0.628989105	0.742739858	0.117802319	0.304770801	0.290186033	0.103046489
MS0004	0.848032036	0.909602109	0.720754118	0.728824909	0.100780465	0.301508979	0.281721183	0.100740817
MS0005	0.802400045	0.913800187	0.751820462	0.731099641	0.118000204	0.300729019	0.27891379	0.104770202
MS0006	0.818070113	0.900010287	0.688770846	0.700071609	0.110017172	0.300040272	0.28801701	0.100017003
MS0007	0.703888177	0.811441408	0.708410482	0.708730786	0.100881386	0.307380013	0.180004124	0.100780120
MS0008	0.881701909	0.903480009	0.781000114	0.734848411	0.114007988	0.308803196	0.278800109	0.100001109
MS0009	0.914140819	0.910000009	0.810100009	0.781000009	0.118000009	0.308800109	0.278800109	0.100770001
MS0010	0.878107010	0.900000007	0.800000007	0.737180005	0.110488878	0.300488878	0.278870109	0.100488878
MS0011	0.888400007	0.940000009	0.781000009	0.731000009	0.100000007	0.307188884	0.278888884	0.100718888
MS0012	0.888414719	0.900000010	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0013	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0014	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0015	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0016	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0017	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0018	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0019	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0020	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0021	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0022	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0023	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0024	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0025	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0026	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0027	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0028	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0029	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0030	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0031	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0032	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0033	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0034	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0035	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0036	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0037	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0038	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0039	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0040	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0041	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888
MS0042	0.888400007	0.900000009	0.781000009	0.734884009	0.118888888	0.308888888	0.278888888	0.100788888



Amazon S3 > Buckets > msba-financialg-999

msba-financialg-999 Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name	Type	Last modified	Size	Storage class
data_sources/	Folder	-	-	-
prediction_file/	Folder	-	-	-



ETL: AWS Glue

AWS Glue Studio [Info](#)

Create job [Info](#)



Author in a visual interface focused on data flow.

Visual ETL



Author using an interactive code notebook.

Notebook



Author code with a script editor.

Script editor

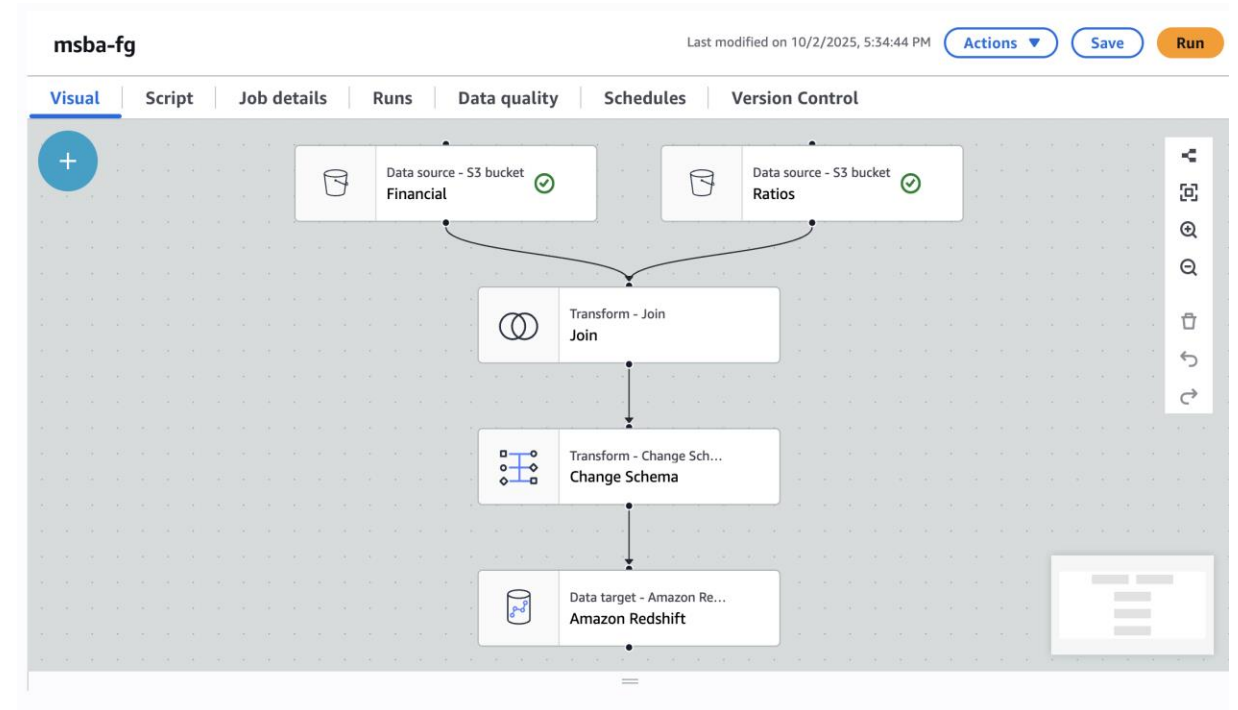
Example jobs [Info](#)

[Create example job](#)

Your jobs (1) [Info](#)

[Refresh](#)
[Actions](#)
[Run job](#)

☐	Job name	Type	Created by	Last modified	AWS Glue version	Action
☐	msba-fg	Glue ETL	Visual	10/2/2025, 5:34:44 PM	5.0	-





Data Warehouse: AWS Redshift

[Amazon Redshift](#) > [Clusters](#) > msba-financialg-cluster-999

msba-financialg-cluster-999

Actions ▾

Edit

Add partner integration

Query data ▾

General information [Info](#)

↻

Cluster identifier msba-financialg-cluster-999	Status ⏸ Paused	Node type ra3.large	Endpoint 📄 msba-financialg-cluster-999.cg2wxytyaxal.us-east-1.redshift.amazonaws.com:5439/dev
Custom domain name -	Date created October 02, 2025, 12:29 CDT	Number of nodes 2	JDBC URL 📄 jdbc:redshift://msba-financialg-cluster-999.cg2wxytyaxal.us-east-1.redshift.amazonaws.com:5439/dev
Cluster namespace ARN 📄 arn:aws:redshift:us-east-1:261976303926:namespace:3d6f4f71-6d01-4a54-a0c4-08b506ef5456	Multi-AZ No	Patch version -	ODBC URL 📄 Driver={Amazon Redshift (x64)}; Server=msba-financialg-cluster-999.cg2wxytyaxal.us-east-1.redshift.amazonaws.com; Database=dev
Namespace register status ⏸ Deregistered	Cluster configuration Production	Storage used 0.00 of 15.3 TB used (0.00%)	



Machine Learning: AWS Sagemaker

Model status [Quick build](#)

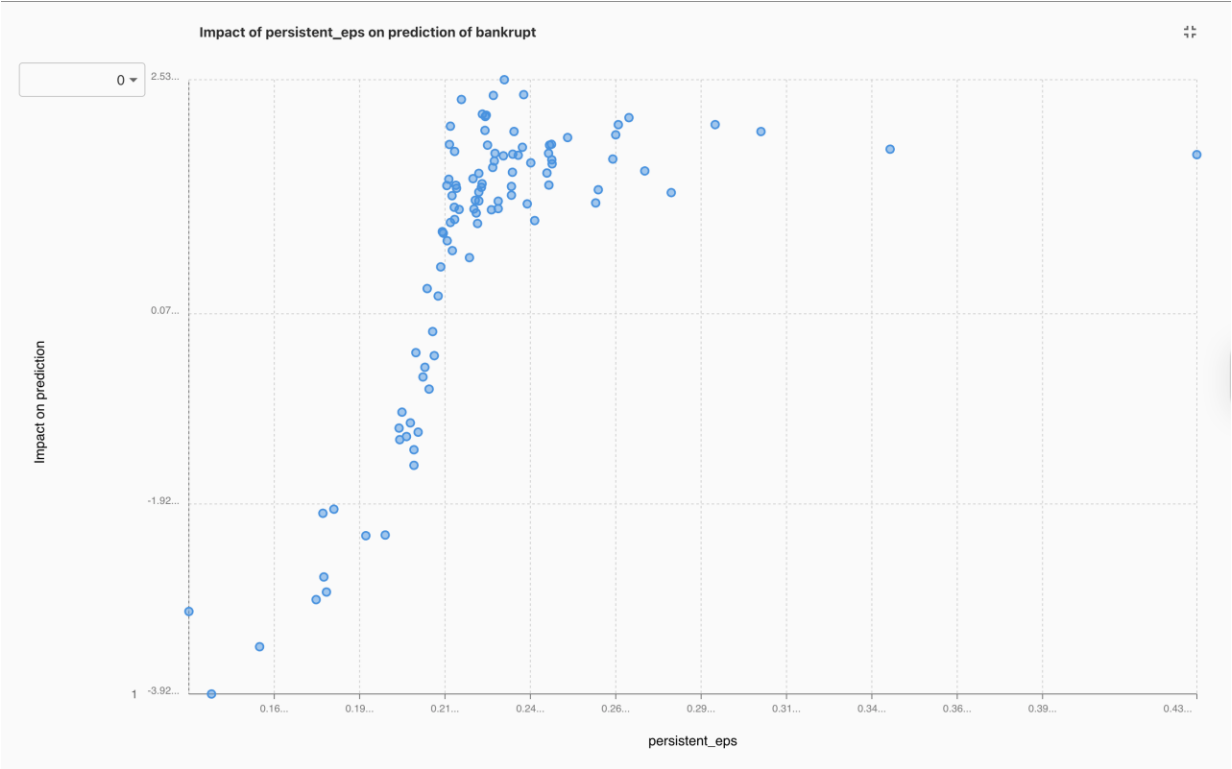
Accuracy ⓘ F1 ⓘ Optimization metric

96.628% 0.489

The model predicts the correct Bankrupt 96.628% of the time. ⓘ

Column impact ⓘ ↓

1	<u>persistent_eps</u>	14.916%
2	<u>borrowing_dependency</u>	8.926%
3	<u>roa_before_interest_percent_after_tax</u>	7.805%
4	<u>operating_profit_paid_in_capital</u>	7.291%



Bankruptcy prediction and recommendation

Invest in:

- Rogers and Sons
- Pharmasolve
- Ninetech

Avoid investing in:

- Western Corp

Prediction (bankrupt)	Probability	borrowing_d...	company	company_id	current_liabi...	current_liabi...	debt_ratio_p...
1	99.7%	0.458818609	western corp	id_6988	0.372217526	0.060766259	0.269038909
0	99.4%	0.37930429	design solutions	id_7413	0.333345174	0.041219716	0.16186474
0	99.2%	0.384998982	innocore	id_8801	0.337392013	0.060765125	0.216101823
0	99.9%	0.374219105	pharmasolve	id_9614	0.329803726	0.030201105	0.108202074
0	99.9%	0.370253398	ninetech	id_9131	0.328092756	0.021710461	0.058590561
0	99.9%	0.37450876	songster inc	id_7102	0.330409488	0.025494302	0.121292741
0	99.9%	0.374179962	rogers and sons	id_7012	0.327484654	0.047166348	0.103576503
0	99.9%	0.373113046	Hallandall ag.	id_9904	0.328042001	0.033711602	0.094385827
0	99.9%	0.377306898	Foster & Kruse	id_6905	0.331114215	0.04351399	0.144662454

EDA Findings

Prioritize:

- Net Income to Total Assets
- Net Profit Before Tax to Paid-in Capital
- Per Share Net Profit Before Tax

Look out for:

- Debt Ratio Percentage
- Current Liabilities to Current Assets

